📘 PROJECT PROPOSAL

StudyBuddy – A Collaborative Study Management Mobile Application

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1️⃣ Introduction

StudyBuddy is a mobile application designed to help students collaborate effectively in study groups, manage academic tasks, and stay organized through shared deadlines and reminders.

Many students struggle with coordinating group work, tracking assignments, and maintaining productive communication. StudyBuddy aims to solve this problem by providing a centralized platform for study collaboration.

2️⃣ Problem Statement

Students face the following challenges:

- Difficulty organizing study groups
- Poor task coordination in group assignments
- Missed deadlines due to lack of reminders
- Inefficient communication across multiple platforms
- No centralized academic collaboration system

Existing solutions either focus only on messaging or only on task management. There is a need for an integrated academic collaboration tool.

3️⃣ Objectives

The main objectives of StudyBuddy are:

1. To provide secure user authentication.
2. To enable students to create and join study groups.
3. To allow group-based task creation and management.
4. To provide deadline reminders via notifications.
5. To support real-time collaboration features (future enhancement).

4️⃣ Proposed Solution

StudyBuddy will be a mobile application built using modern technologies that support scalability, security, and performance.

The system will:

- Authenticate users securely
- Allow creation and management of study groups
- Support shared task tracking within groups
- Provide notification reminders before deadlines
- Store all academic collaboration data securely in the cloud

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5️⃣ System Architecture

Frontend Technologies

- React Native (Expo Framework)
- Expo Router for navigation
- NativeWind for styling
- Zustand for state management
- react-hook-form for form handling
- axios for API communication
- expo-notifications for reminder scheduling

Backend Technologies

- Firebase for authentication
- Supabase for relational database storage

This architecture separates authentication from database management, ensuring scalability, modularity, and maintainability.

6️⃣ Database Design Overview

The application will include the following relational tables:

- Users

- Study Groups
- Group Members
- Tasks
- Messages (future enhancement)

The relational structure ensures proper linkage between users, groups, and tasks, allowing efficient data retrieval and consistency.

7️⃣ Key Features

Phase 1 – Core Features

- User registration and login
- Profile creation
- Create and join study groups
- View group list

Phase 2 – Task Management

- Create group tasks
- Set deadlines
- Mark tasks as completed
- View tasks by group

Phase 3 – Notifications

- Local notifications for deadlines
- Reminder scheduling

Phase 4 – Future Enhancements

- Real-time group chat
- File sharing
- Role-based permissions (Admin/Member)
- Dark mode support

8️⃣ User Interface & Design System

StudyBuddy will implement a structured design system to ensure consistency, accessibility, and scalability across light and dark modes.

🎨 Brand Identity

The application's primary brand color is Jacksons Purple (HSL 239°, 70%, 34%), which represents focus, intelligence, and academic calmness.

A full color scale (100–800 variations) will be derived from this base color to support hover states, active states, and accessibility adjustments.

🌞 Light Mode Design

Light Mode will utilize:

- Background: Very light neutral (HSL 0°, 0%, 98%)
- Surface/Card Background: Soft neutral (HSL 0°, 0%, 95%)
- Primary Text: Dark neutral (HSL 0°, 0%, 9%)
- Secondary Text: Medium neutral shades
- Primary Buttons: Jacksons Purple (500 scale)
- Secondary Buttons: Light purple variations

This ensures clarity, high readability, and a clean academic interface.

🌚 Dark Mode Design

Dark Mode will utilize:

- Background: Dark neutral (HSL 0°, 0%, 9%)
- Surface: Slightly elevated dark layers
- Primary Text: Light neutral (HSL 0°, 0%, 95%)
- Accent Purple: Slightly brighter variation for visibility

The dark theme will reduce eye strain and improve usability in low-light environments.

🔔 Semantic Colors

The system will also include:

- Success (Green tone)
- Warning (Amber tone)
- Error (Red tone)
- Informational (Blue tone)

These will be used consistently for alerts, validation messages, and system feedback.

📐 Design Principles

- Minimalist and academic aesthetic
- Clear typography hierarchy
- Consistent spacing and elevation system
- Accessibility-compliant color contrast
- Scalable design tokens for maintainability

This structured design system ensures StudyBuddy maintains a professional, modern appearance while supporting long-term feature expansion.

9️⃣ Expected Outcomes

Upon completion, StudyBuddy will:

- Improve student collaboration
- Reduce missed deadlines
- Provide structured academic coordination
- Demonstrate modern full-stack mobile development skills
- Showcase implementation of scalable UI design systems

🔟 Conclusion

StudyBuddy aims to provide a structured, collaborative academic environment for students through a secure and scalable mobile application.

The project demonstrates the integration of modern frontend and backend technologies, structured database architecture, and a professional design system while addressing a real-world academic problem.